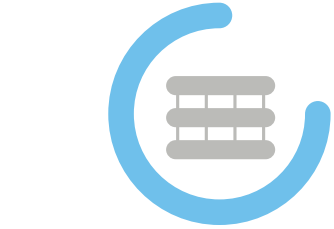

Meeting of the Technical Steering Committee (TSC) Board

Wednesday, April 22nd, 2020
11:00am ET

Meeting Logistics

- https://zoom.us/j/556149142

- United States : +1 (646) 558-8656
  - Meeting ID: 556 149 142

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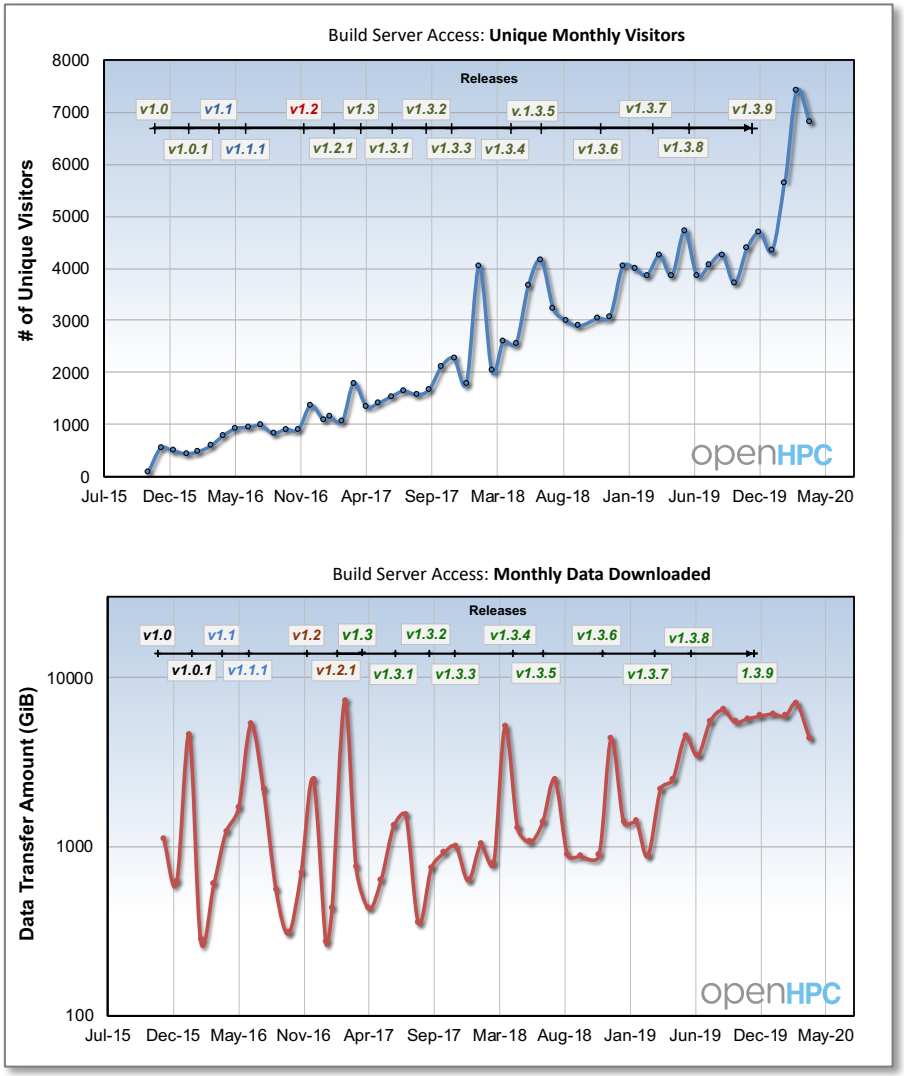
THE LINUX FOUNDATION

Agenda/Updates

- Announcements:
  - ISC in-person event cancelled – will be a digital event instead
    - still have not heard anything yet regarding our accepted BoF
- Upcoming deadlines:
  - SC'20
    - Tutorials?: Due ~~April 16, 2020~~, Extended to April 30 – *Adrian leading a submission)
    - BoFs: Due July 31, 2020
- ARM compiler build work back on track

- Quarterly access stats
- PEARC'20 tutorial/cloud working group updates (csim)
- Netdata eval (csim)
- Additional component deprecation
- Component Review #9 Results
- 2.0 build status

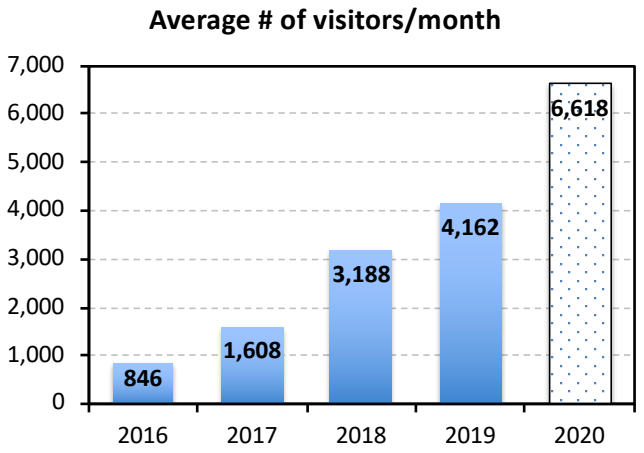
Updated Usage/Access Statistics (thru Q1 2020)



- Stats for build/repo server (tracking # of unique visitors per month and amount of data downloaded):

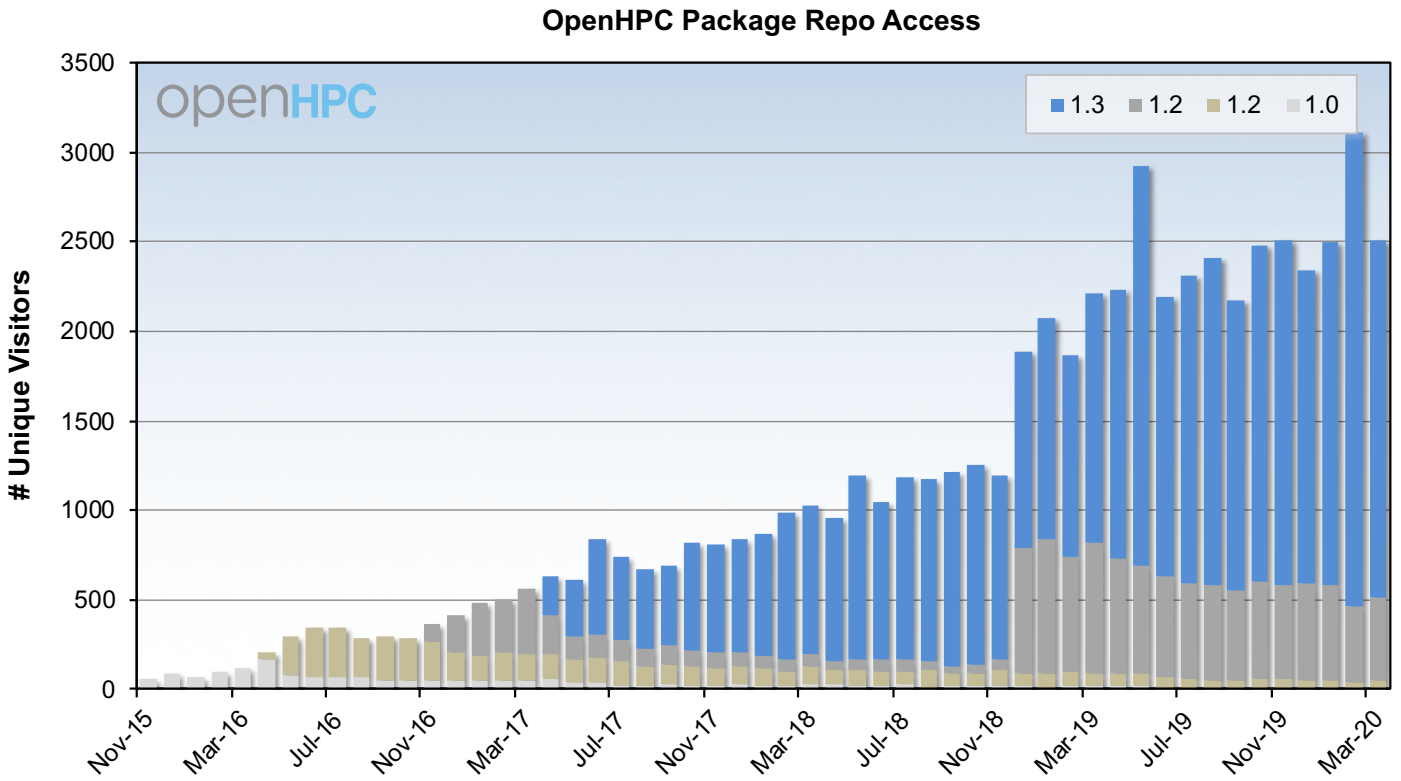
  http://build.openhpc.community

- Averaging ~5.7 TB/month download in 2020



Updated Usage/Access Stats (thru Q1 2020)

- These stats monitor access specifically to package repository metadata (typically expected to be via yum/zypper)
- Repo access binned by minor version



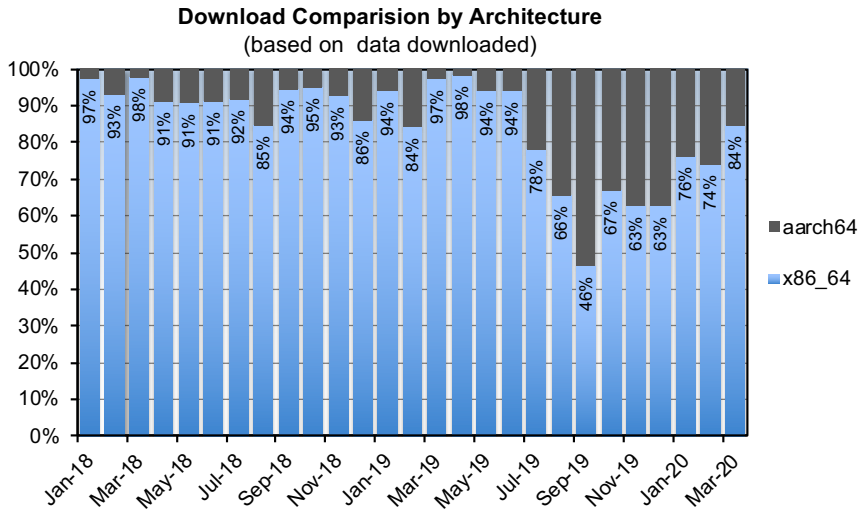
Updated Usage/Access Stats (thru Q1 2020)

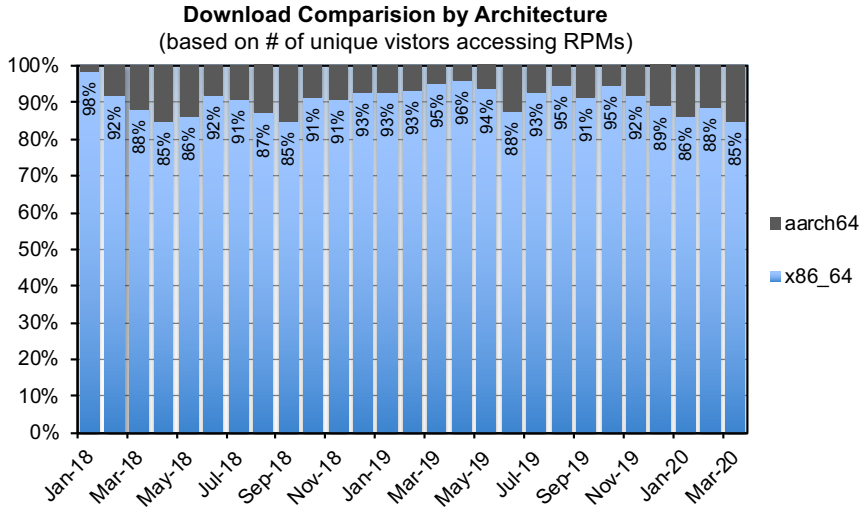
Architecture specific metrics:

- To provide some characterization, we scrape the access logs to analyze two architecture specific file types:

  ```
  (aarch64|x86_64).rpm
  (aarch64|x86_64).tar
  ```

- Plots compare percentages for the amount of data xfer'ed and the # of unique visitors accessing the (aarch64|x86_64) files





OHPC Cloud Working Group Updates (csim)

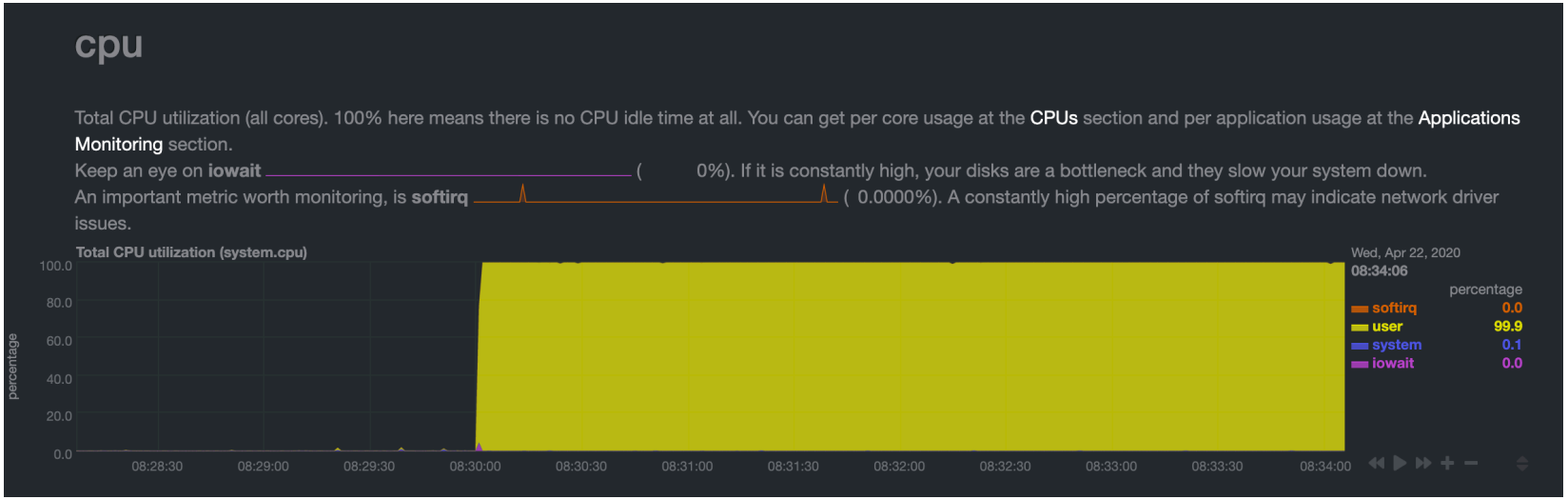
- PEARC is now confirmed to be a virtual event
- Was no cloud working group meeting this week
- Private GitHub repo for project was moved to the OHPC organizational GitHub account
- Let us know if you want access
- Have successfully deployed the “full cluster” version
- Beginning to work more on documentation and getting the OHPC version finished up

Netdata Eval/Overview (csim)

- A GitHub request for the inclusion of Netdata as a monitoring solution for OHPC was suggested
- Ganglia's last release was 2014?
- Netdata is:
  - Real-time health monitoring and performance troubleshooting solution
  - Not a drop in replacement for Ganglia
  - More distributed by design but can be used in a client / server mode with configuration
  - … AMAZING

Netdata cont.

- Very lightweight
  - 3-5% CPU of a single core when sampling every second
  - Half that when collecting only (more on that in a bit)
  - 40-80 MB of RAM used depending on settings
- Amazingly intuitive interface



Netdata basic concepts

- Each node can either have a database of results locally or not
- If database is set to none, the node is a collector only; You can then set it up it in a client/server configuration.
- If a database is configured, a web portal is also run on that node to display the results. Default is port 19999
- Database types:
  - RAM – monitoring data is stored in RAM only; ~100 MB will give you about 6 hours of backlogs depending on the number of collectors that get configured
  - Dbengine – RAM + disk; most recent data is in RAM and older data gets written to disk; Web UI has access to all data just a little slower accessing older data

What netdata doesn't do today

- At this time, it doesn't provide "cluster views"
- It has no concept of a larger machine
- It is not easy to compare metrics across multiple nodes
- Netdata Cloud (releasing next week) might provide some solutions
- However, Netdata Cloud will require using their cloud service initially

BigTex Grid (1 sources) (tree view)

CPUs Total:	**13648**
Hosts up:	**850**
Hosts down:	**0**

Longer Term Storage and Visualization

- Any netdata node can also save data to a backend
- Many backends supported
  - Graphite, InfluxDB, KairosDB, Blueflood, ElasticSearch
  - OpenTSDB, InfluxDB, KairosDB, MongoDB
  - json document DBs, Prometheus, TimescaleDB
- Only one backend per node at a time but you can chain together multiple netdata nodes with different backends
- Netdata can down sample the data to reduce storage at the cost of fidelity
- At UTD, we're moving forward with netdata + graphite + grafana

Netdata + Graphite and Grafana

- Netdata runs on every cluster node and forwards those results to a master netdata server
- No local database is used on compute nodes
- A netdata master server will down sample data to 10 second intervals
- Store all cluster node netdata data in a Graphite database
- Graphite web front end can then be used to make multi-host "charts on demand" to assist with debugging
- Grafana will read from the Graphite database (and other data stores)
- Grafana dashboards can be written to integrate Netdata and Queue system logs/accounting to create new capabilities
- Goal is to have a historical job dashboard where clicking on a 'jobid' gives you access to all of the netdata metrics during the lifetime of your job and a platform to analyze the data

Netdata worth considering for OHPC?

Thoughts? Questions?

Additional component deprecation

- We discussed deprecation of mpiP/ocr last time
- Would like to discuss doing the same for one additional package for which execution issues have cropped up in CI:
  - mrsh
    - provides functionality like ssh/scp, but uses **munge** for underlying authentication
    - not part of critical path in current installation recipes (optional)
    - last release was April 2016
    - relies on tcp_wrappers library which was deprecated in rhel7 and not available in rhel8 (https://access.redhat.com/solutions/3906701)

Component Review #9 Results

Component Name	# of Reviewers	# of accepts	# of rejects	Avg. Priority
Libfabric	6	4	2	7.0
OpenFlightHPC User Suite	7	1	5	2.2
LLNL Magpie	5	5	0	6.3

- Discussion on libfabric:
  - those who voted to reject were doing so in favor of using the distro provided version (1.8.x available in both Leap and CentOS8)
  - i'm also inclined to do that but update MPI builds accordingly and document/test installation in recipes

2.0 Build Status

- Have enabled a good number of packages in 2.0 Factory
  - introduction of arm1 compiler variant means 10 compiler/MPI permutations for MPI-based builds

Administrative Tools

Package	Built?	Notes
conman	✓	
docs	✓	
examples	✓	
~~ganglia~~		
genders	✓	
lmod-defaults	✓	added (3/10/20)
losf	✓	
mrsh	✓	fixed (3/8/20)
~~nagios~~		
~~nagios-plugins~~		
~~ndoutils~~		
nhc	✓	added (3/10/20)
~~nrpe~~		
pdsh	✓	
prun	✓	
test-suite	✓	

Compiler Families

Package	Built?	Notes
gcc9	✓	
intel-compatibility	✓	
arm-compatibility	✓	
llvm		

Development Tools

Package	Built?	Notes
easybuild	✓	
autoconf	✓	
automake	✓	
cmake	✓	
hwloc	✓	
libtool	✓	
python-mpi4py	※	3/24/20 – arm1 failures
python-numpy	※	3/24/20 – arm1 failures
python-scipy	※	3/19/20 – arm1 failures
spack	✓	[still targets root usage though]
valgrind	✓	

MPI Families

Package	Built?	Notes
impi-compatibility	✓	no longer requires install of intel compat. package
mpich	✓	
mvapich2	✓	
openmpi4	✓	

Serial Libs

Package	Built?	Notes
R	✓	
GSL	✓	
metis	✓	
openblas	✓	
plasma	✓	arm1 enabled (4/21/20)
superlu	✓	arm1 fixed (3/9/20)

※ = partial builds success
✓ = all builds complete
X = all builds fail

2.0 Build Status (cont.)

Parallel Libraries

Package	Built?	Notes
boost	※	intel error
fftw	✓	build fixed
hypre	※	intel/arm1 blas issue
mfem	※	needs deps
mumps	※	arm1/impi issue
opencoarrays	※	leap/openmpi4 failures
petsc	※	intel/arm1 issue
scotch	✓	arm1 fixed (3/9/20)
scalapack	✓	arm1 fixed (3/9/20)
slepc	※	waiting on petsc deps
superlu_dist	※	2 Leap 15.1 failures
trilinos	※	Intel/ARM failures (4/8/20)

Resource Management

Package	Built?	Notes
PBS Pro	※	need libical-devel on aarch/centos
pmix	✓	
slurm	✓	

Performance Tools

Package	Built?	Notes
dimemas	※	need some boost builds, suse failures
extrae	※	arm1 failures
geopm	※	intel failures (4/18/20)
imb	※	updated to v2019.6, all built except for arm1 failures (4/16/20)
likwid	✓	(4/7/20)
msr-safe	✓	(4/20/20)
~~mpiP~~		
omb	✓	3/4/20
papi	✓	
paraver	✓	added 3/5/20
pdtoolkit	※	intel/leap failure (3/5/20)
scalasca	※	need scorep deps (3/5/20)
scorep	※	variety of issues (3/5/20)
tau	※	need pdtoolkit on intel/leap and arm1 (3/24/20)

Runtimes

Package	Built?	Notes
charliecloud	✓	3/23/20
~~ocr~~		
singularity	✓	updated to use newer go version (4/6/20)

※ = partial builds success
✓ = all builds complete
X = all builds fail

2.0 Build Status (cont.)

I/O Libraries

Package	Built?	Notes
adios	※	intel failures (and deps)
hdf5	※	Leap/arm1 failures
netcdf-cxx	※	Leap/arm1 failures
netcdf-fortran	※	Leap/arm1 failures
phdf5	※	arm1 failures
pnetcdf	✓	
sionlib	※	arm1 failures

Provisioning

Package	Built?	Notes
warewulf	✓	starting to run in CI (3/10/20)

※ = partial builds success
✓ = all builds complete
X = all builds fail

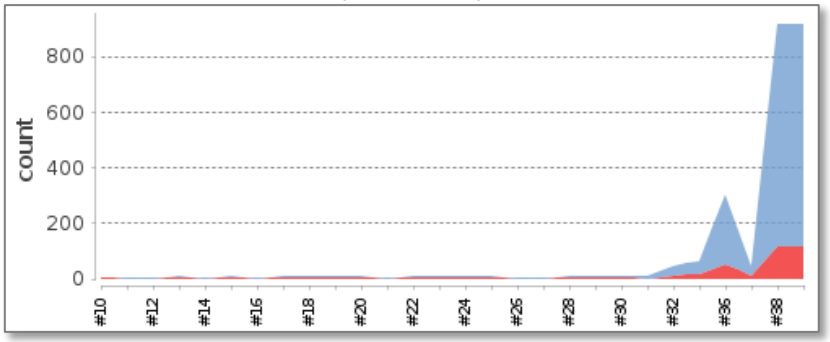
2.0 Build Status (cont)

- Current package counts:
  - ~~800 RPMs as of 3/11/2020~~
  - ~~905 RPMs as of 3/25/2020~~
  - ~~982 RPMs as of 4/8/2020~~
  - 1003 RPMs as of 4/22/2020

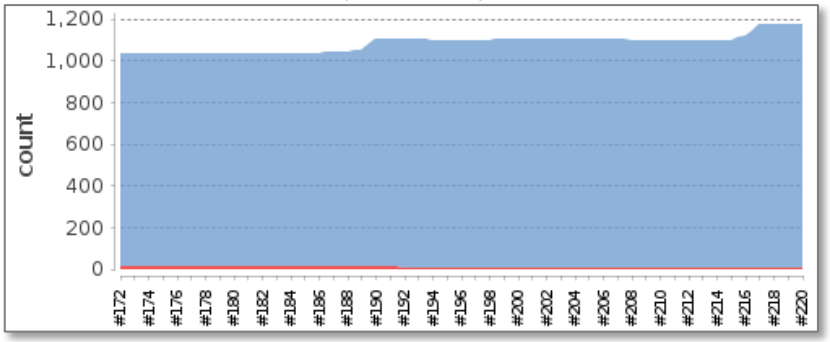
Base OS	aarch64	x86_64	noarch
CentOS 8	165	316	29
Leap 15	162	302	29

2.0 (cont.) – CI updates

centos8.1/x86/WW/slurm (3/10/2020)



centos8.1/x86/WW/slurm (4/22/2020)



- From last time (3/10):
  - ~~610 user level tests passing~~
- As of 3/25:
  - ~~910 user level tests passing~~
- As of this morning (4/08):
  - ~~982 user level tests passing~~
- As of this morning (4/22):
  - 1,132 user-level tests passing

2.0 (cont.) – CI updates

add description

2.0 | All | +

S	Categorized - Job	Last Success	Last Failure	Last Duration	Test Result
	.. » [aarch64]	18 days - #3	4 hr 9 min - #1	7 min 57 sec	N/A
	(2.0) - (centos8.1,aarch64) (warewulf+slurm) (fabric=eth)	19 days - #47	13 hr - #54	29 min	5 of 1,103 failed (-9)
	(2.0) - (leap15.1,aarch64) (warewulf+pbspro) (fabric=eth)	N/A	4 hr 9 min - #1	1 hr 1 min	14 of 800 failed (+14)
	(2.0) - (leap15.1,aarch64) (warewulf+slurm) (fabric=eth)	18 days - #3	5 days 20 hr - #10	7 min 57 sec	13 of 808 failed (-13)
	.. » [x86_64] - CentOS 8	28 days - #3	1 hr 31 min - #222	3 min 14 sec	N/A
	(2.0) - (centos8.1,x86_64) (warewulf+slurm) (fabric=eth)	1 mo 13 days - #27	1 hr 31 min - #222	3 min 17 sec	3 of 1,174 failed (-4)
	(2.0) - (centos8.1,x86_64) (warewulf+slurm) (fabric=ib) + psxe	28 days - #3	3 hr 49 min - #50	3 min 14 sec	5 of 1,210 failed (-14)
	.. » [x86_64] - Leap15	28 days - #9	59 min - #295	3 min 37 sec	N/A
	(2.0) - (leap15.1,x86_64) (warewulf+pbspro) (fabric=eth)	N/A	3 hr 39 min - #5	38 min	12 of 841 failed (±0)
	(2.0) - (leap15.1,x86_64) (warewulf+slurm) (fabric=eth)	1 mo 6 days - #6	59 min - #295	3 min 48 sec	12 of 1,014 failed (±0)
	(2.0) - (leap15.1,x86_64) (warewulf+slurm) (fabric=ib)	28 days - #9	28 days - #15	3 min 37 sec	35 of 823 failed (±0)

v1.3.9 totals
1,166 tests
1,156 tests
1,166 tests
1,237 tests
2,510 tests
1,203 tests
1,549 tests

- Enabled some pbspro jobs on Leap
- We are unfortunately down 1 of our Jenkins aarch64 nodes in the UK
  - host throwing h/w error during bios post

Moving forward with a preview release

- Reasonably content with ethernet only CI tests
- Propose to:
  - finalize MPI builds with libfabric (and UCX where applicable)
  - agree/enable on default build flags
  - achieve similar/reasonable test coverage on InfiniBand runs
  - work thru an updated release and test process as the public packages will be pushed to repos.openhpc.community
  - push out a subset of recipes (but coverage for both Leap and CentOS)
    - Warewulf only
    - No arm1 compiler coverage initially
  - document known issues and packages currently unavailable